



LEGAL METROLOGY (PACKAGED COMMODITIES)
RULES, 2011 · COMPLIANCE AUTOMATION

NEXT.JS · NODE.JS · MONGODB

EASYOCR LABEL READING

ENFORCEMENT DASHBOARD

LM-Verify

Automated compliance checking

for packaged commodities

A web application that scans package labels through the device camera, extracts every mandatory declaration with OCR, tests it against the Legal Metrology (Packaged Commodities) Rules, 2011, and issues a downloadable compliance report.

Document

Key Features & Workflow

Version

1.1 — Solution Design

Stack

Next.js · Node.js · MongoDB

Date

14 September 2026

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What this document covers

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SECTION 01

Context & objective

Every pre-packaged commodity sold in India must carry the declarations listed in **Rule 6** — manufacturer / packer / importer identity, the common or generic name, net quantity, month and year of manufacture or packing, retail sale price, and consumer care contact details — each rendered in a prescribed size, place and manner under **Rules 7 to 12**.

Verification today is manual. An inspector reads the panel, measures numerals against a scale, compares wording to the statute, and writes the finding by hand. Against a market of millions of SKUs across physical retail and e-commerce, coverage stays thin and inconsistent between officers.

The objective. LM-Verify turns that manual reading into a repeatable machine process. A photograph of the package is conditioned, read by OCR, normalised into structured fields and tested against a versioned rule pack. Each rule returns an explicit verdict carrying a citation and a cropped image as evidence.

The officer remains the decision-maker. The system does the reading, the measuring and the paperwork; the officer confirms, overrides where needed, and downloads the signed report.

6	28	2	₹2k–4k
DECLARATION HEADS	AUTOMATED CHECKS	FONT-HEIGHT TABLES	PENALTY PER BREACH

SECTION 02

System architecture

A Next.js front end, a Node.js API, MongoDB for persistence, and one Python sidecar that runs EasyOCR. All statutory judgement happens in a single deterministic rule service, so any verdict can be reproduced and defended later.

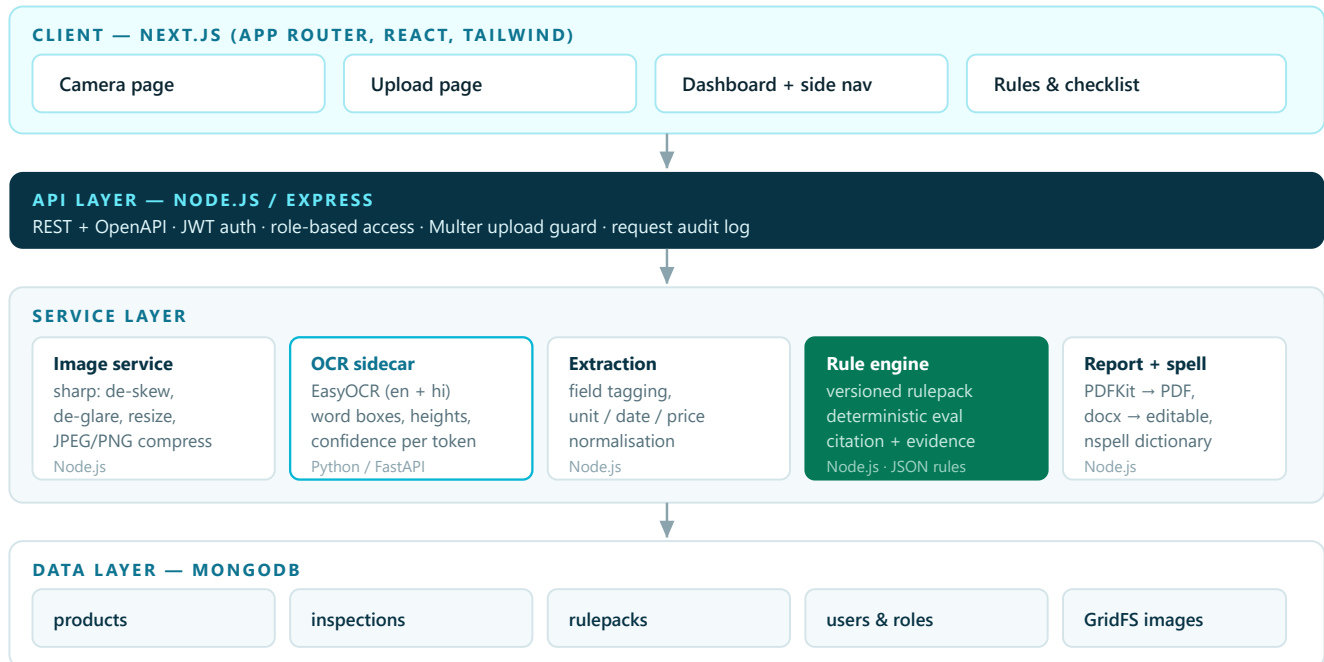


Figure 1 Layered architecture. Arrows show request flow; each service scales independently behind the Node API.

WHY A PYTHON SIDECAR

EasyOCR is a Python library. It runs as a small FastAPI service that Node calls over HTTP, returning word boxes, pixel heights and per-token confidence. The rest of the system stays pure JavaScript.

WHY MONGODB

A label is a ragged document — fields present or absent, arrays of violations, nested evidence crops. A document store fits it without schema gymnastics, and text indexes power the search screen.

DETERMINISTIC VERDICTS

OCR reads the label. It never decides compliance. Verdicts come from a JSON rule pack a human can open, read, diff and version — which is what makes a finding defensible.

SECTION 03

The processing pipeline

What happens between the shutter click and the verdict. Each stage hands a richer object to the next; every stage writes its own confidence score, and low confidence routes the case to a human rather than guessing.

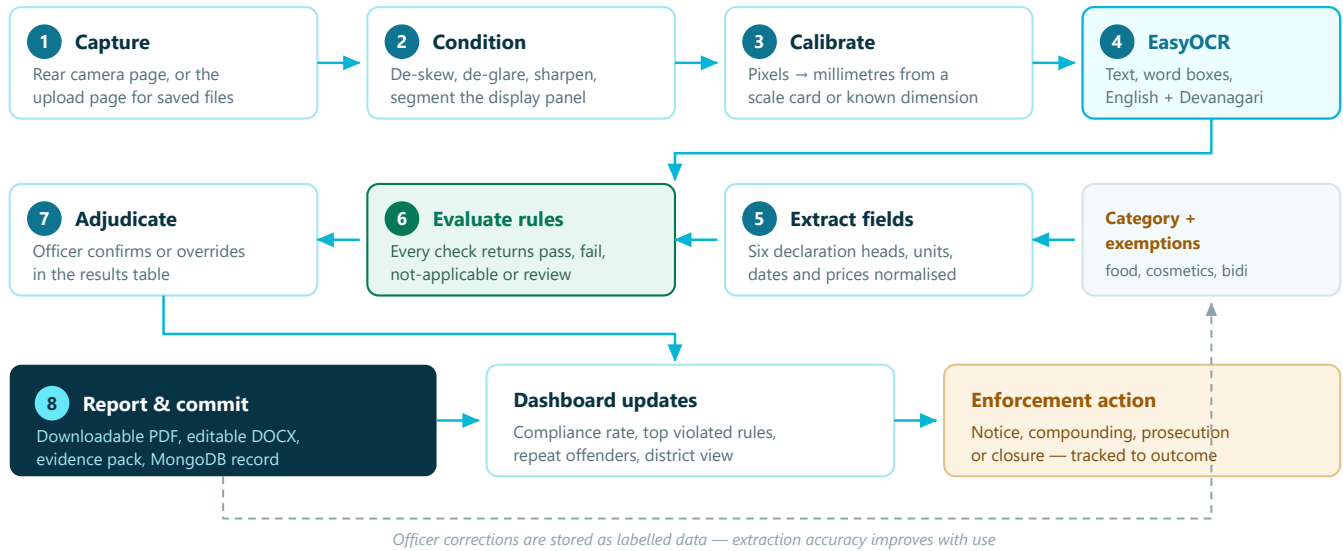


Figure 2 The eight-stage pipeline. Stages 1–6 are automatic; stage 7 is always human.

Why stage 3 exists. Rule 7(2) prescribes minimum numeral heights in millimetres — 1 mm, 2 mm, 4 mm and 6 mm depending on net quantity and panel area. A photograph carries no physical scale, so font-height compliance cannot be judged from pixels alone. The capture flow therefore requires a reference scale card in frame, or a known package dimension typed in by the officer. Without one, the font-size checks return **needs review** instead of a false verdict.

SECTION 04

Application map & interface

A persistent side navigation bar anchors the whole application. Capture and upload are deliberately separate destinations rather than two tabs on one screen — each has its own guidance, its own validation and its own failure states.

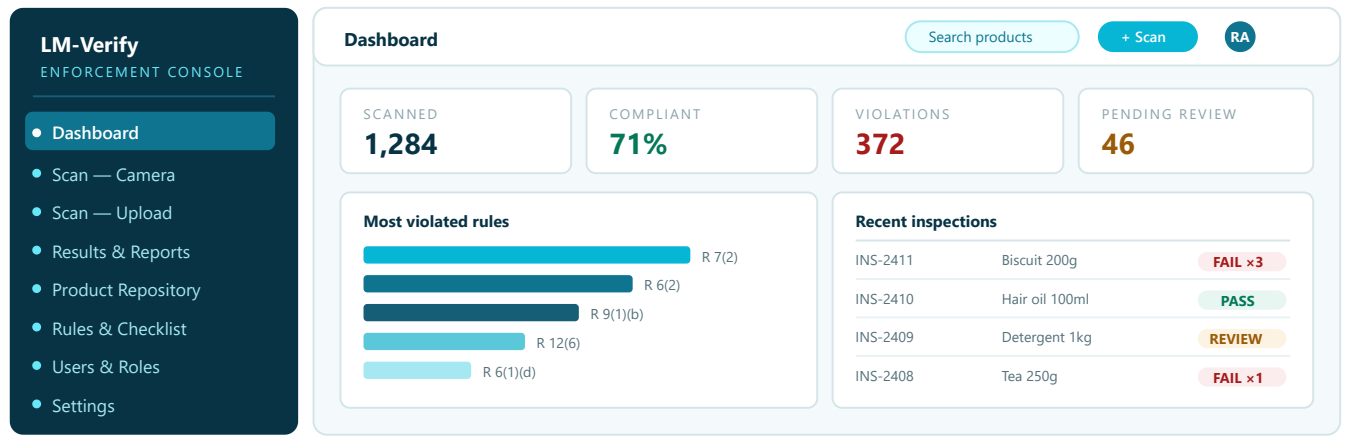


Figure 3 Dashboard with the persistent side navigation. The same rail appears on every screen.

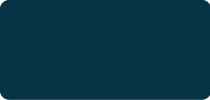
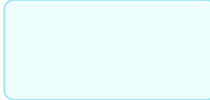
ROUTE	SCREEN	WHAT IT DOES
/dashboard	Dashboard	KPI tiles, most-violated-rules chart, district breakdown, recent inspections feed, pending-review queue.
/scan/camera	Camera capture	Live rear-camera viewfinder with a panel-alignment frame, torch toggle, multi-shot capture for each package face, blur and glare warnings before the shot is accepted.
/scan/upload	Image upload	Drag-and-drop or file picker for JPEG / PNG, single or batch, with format and size validation and a thumbnail tray before submission.
/results/[id]	Extraction & verdict	Two tables side by side — extracted text and the compliance checklist — plus the annotated image, spell-check panel and the download buttons.
/repository	Product repository	Every scanned product with its compliance history, searchable by brand, category, rule violated, officer, district or date.
/rules	Rules & checklist	The full rule set rendered as a reference table, with the live checklist state for the product currently open.
/admin/users	Users & roles	Account creation, role assignment, jurisdiction mapping, audit trail of privilege changes.

SECTION 05

The cyan design system

One accent family, used with discipline. Cyan carries identity and interaction; verdict colours are reserved strictly for compliance status so that a red cell always means a violation and never merely decoration.

Palette

						
Deep #083344 Nav rail, headings	Brand #0E7490 Links, active state	Bright #06B6D4 Primary buttons	Soft #ECFEFF Panels, chips	Pass #047857	Review #9A5B08	Fail #A61B1B

TYPE

A single geometric sans across the interface — Inter or the system UI stack. Numerals are set with tabular figures so quantity, price and millimetre columns align down the table.

SURFACE

Near-white page, white cards, one-pixel cyan-tinted borders instead of drop shadows. Depth comes from the border and a faint tint, which keeps long tables readable.

MOTION

Restrained. A progress bar during OCR, a skeleton row while a table loads, a soft pulse on the capture button. Nothing animates that the officer has to wait for.

Interface rules

- **Side rail, always.** Fixed 240 px navigation on desktop; it collapses to icons on tablet and becomes a bottom sheet on phones. The active item is filled in brand cyan.
- **Status is a colour, not a word alone.** Every verdict cell carries both a coloured pill and its text label, so the table survives greyscale printing and colour-blind readers.
- **Tables over cards for data.** Extraction output and compliance results are tabular by nature; cards are reserved for KPI tiles and empty states.
- **One primary action per screen.** On the camera page it is the shutter; on upload it is submit; on results it is download report. Everything else is secondary or ghost styling.
- **Dark mode ships from day one.** Tokens are declared once; the deep cyan ground inverts cleanly, which matters for godown and warehouse lighting.
- **Accessibility floor.** Text contrast at 4.5:1 or better, keyboard-reachable table controls, and visible focus rings in bright cyan.

Why cyan works here. It reads as instrument-panel and measurement rather than as alarm. That leaves the warning register — amber and red — entirely free for compliance meaning, which is the one thing in this application that must never be ambiguous.

SECTION 06

Capture & extraction features

MODULE A

GETTING THE IMAGE IN

1 Rear-camera capture page DEDICATED PAGE

A live viewfinder pinned to the device's back camera via `getUserMedia({ facingMode: "environment" })`, falling back to the default camera where the constraint is unsupported. An alignment frame shows where the principal display panel should sit, a torch toggle handles dim shelves, and blur or glare is caught before the frame is accepted rather than after.

2 Separate image upload page DEDICATED PAGE

Its own route for images already on the device — drag-and-drop or file picker, single or batch. Format and size are validated client-side and thumbnails appear in a tray. Keeping this apart from the camera page means neither flow carries the other's error states.

3 Attachment policy — JPEG / PNG, 10–15 KB

Supporting photographs accept only `image/jpeg` and `image/png`, enforced by MIME sniffing on the server rather than by file extension. Each is compressed to the 10–15 KB envelope by `sharp` on upload, so reports stay small and download quickly on field connections.

4 Image conditioning before OCR

Perspective correction, de-skew, glare suppression, contrast normalisation and upscaling, then panel segmentation separating the principal display panel from side and back panels — the distinction Rule 8 turns on.

5 Millimetre calibration

A reference scale card in frame, or a known package dimension entered by the officer, converts pixels to millimetres. This is what makes the Rule 7 font-height check a measurement rather than a guess.

One note on the size limit. A 10–15 KB JPEG is heavily compressed — fine as a report attachment, too lossy for OCR or evidentiary zoom. So extraction runs on the full-resolution capture, that original is kept in GridFS against the inspection, and only the copy travelling inside the report is compressed. Nothing is measured from the compressed image.

MODULE B

READING THE LABEL

6 EasyOCR text recognition

EasyOCR runs in a Python service and returns, for every word, the recognised string, its bounding box and a confidence score. Bilingual by default — English and Devanagari — as Rule 9(4) allows declarations in Hindi, English or both. Box heights feed straight into the font-size analyser.

7 Raw extraction table TABLE VIEW

The first of two tables on the results screen. Every OCR token is listed with its text, position, confidence and the panel it sat on. Low-confidence rows are tinted amber and directly editable, so a misread character is corrected without rescanning.

8 Structured declaration table TABLE VIEW

The second maps raw text onto the six statutory heads — manufacturer, generic name, net quantity, month and year, MRP, consumer care — showing declared value, normalised value and source crop side by side. Both tables export to XLSX.

9 Normalisation

Units canonicalised as Rule 13 prescribes; prices parsed with the rounding convention in Rule 2(m); dates accepted as words, numerals or both, per the Explanation to Rule 6(1)(d).

SECTION 07

Rule engine & analysis

Module C is the compliance core. Rules live as versioned JSON in MongoDB rather than as code, so an amendment is a data change reviewed by a legal officer — not a release.

10 Declarative rule pack

Each rule carries an id, the statutory citation, an applicability condition, a predicate, a severity and the penalty exposure. Every evaluation records which rule-pack version produced it, so a finding from last year can still be reproduced exactly.

11 Mandatory declaration completeness Rule 6

Presence and non-emptiness of all six heads, plus the consumer care contact required by Rule 6(2) — name, address, telephone and e-mail where available. Missing heads are reported individually, never as one lumped failure.

12 Font height and width analysis Rule 7(2)–(3)

Measured glyph height in millimetres is tested against Table I for quantities by weight or volume (1 mm up to 200 g/ml, 2 mm above 200 up to 500, 4 mm above 500) and Table II for length, area or number, which keys off principal-display-panel area (1, 2, 4 and 6 mm bands). Blown, formed, moulded, embossed or perforated containers are held to the doubled figures. Letter width is separately checked at no less than one third of height.

13 Placement and clear space Rule 8

Confirms declarations appear on the principal display panel, and that the quantity declaration is surrounded by clear space — at least one numeral height above and below, and twice the numeral height left and right — measured geometrically from the OCR boxes.

14 Legibility and contrast Rule 9(1)

Luminance contrast between MRP and net-quantity numerals and their local background is computed and flagged where the colour does not contrast conspicuously. Read-through-liquid cases under Rule 9(2) are flagged for manual confirmation.

15 MRP form and sticker misuse Rules 2(m), 6(3)

Checks the prescribed wording — **Maximum Retail Price Rs ... inclusive of all taxes** or the abbreviated form — and detects stickers pasted over a printed MRP, which Rule 6(3) permits only for a downward revision and never covering the original declaration.

16 Misleading quantity language Rule 12(6)

Flags qualifiers near the quantity declaration — *minimum, not less than, average, about, approximately* and similar expressions — which the rule expressly forbids, along with *when packed* outside the Third Schedule cases.

17 Spell check on product description NEW

Descriptive text captured from the label or typed into the inspection form is run through a dictionary checker (`nspell` with an Indian-English word list, extended by a curated domain lexicon of brand names, commodity terms and statutory phrases). Misspellings are underlined inline with suggested corrections, and the checker doubles as a misread detector: a cluster of nonsense words usually means poor OCR on that region, which the system surfaces as a rescan prompt rather than a violation.

18 Category and exemption resolution

Food articles defer to FSSAI labelling, cosmetics to the Drugs and Cosmetics Rules, and the carve-outs for bidi, incense sticks and domestic LPG cylinders suppress the relevant checks. Suppressed rules are shown as **NOT APPLICABLE** with the reason, never silently hidden.

SECTION 08

Reports, repository & dashboards

MODULE D

OUTPUT

19 Downloadable compliance / non-compliance report PDF

One button on the results screen produces the report. A compliant product yields a certificate of compliance; a non-compliant one yields a violation report listing each failed check, its rule citation, the measured value against the prescribed value, and the cropped evidence image. Both are generated server-side and stream straight to the browser as a download.

20 Editable export formats

The same report exports as DOCX for officers who must edit wording before issue, and the underlying tables export as XLSX for bulk analysis. PDF for the record, DOCX for the draft, XLSX for the numbers.

21 Evidence pack and chain of custody

Original capture, annotated overlay, compressed attachments, capture timestamp, geo-coordinates, device fingerprint, officer identity and a SHA-256 hash of each file — bundled so the report can stand up as evidence rather than as a printout.

22 Verification QR

Each issued report carries a QR code resolving to a read-only verification page, so a recipient can confirm the report is genuine and unaltered without an account.

MODULE E

MEMORY & OVERSIGHT

23 Product repository

Every scan is retained against a product identity resolved by barcode or GTIN where present, and by perceptual image hash plus brand and quantity where it is not. Rescanning the same SKU appends to its history instead of creating a duplicate, which is what makes repeat-offender detection possible.

24 Search and retrieval

MongoDB text indexes over brand, manufacturer, generic name and report text, combined with faceted filters for rule violated, verdict, officer, district, category and date range. Any past inspection and its report are retrievable in a few keystrokes.

25 Enforcement dashboards

Compliance rate over time, the most frequently violated rules, repeat-offender brands and manufacturers, a district-level view, officer throughput, and a comparison of physical retail against e-commerce listings. Every chart drills through to the underlying inspection list.

26 Pending-review queue

Cases the engine could not decide with confidence — uncalibrated images, low OCR confidence, ambiguous categories — collect in one queue rather than scattering across the system, so nothing silently passes for want of a decision.

REPORT ANATOMY

Header and inspection reference · product identity · images · declaration table · checklist with verdicts · violation detail with citations · penalty exposure · officer remarks · signature block · verification QR.

PENALTY EXPOSURE

Computed from Rule 32 — ₹4,000 for a contravention of Rules 27 to 31, ₹2,000 for any other contravention not otherwise punishable — shown as an indicative figure, not an adjudication.

BULK MODE

A ZIP of images, or a list of listing URLs, runs through a BullMQ queue and returns a consolidated XLSX plus one PDF per product — a market survey in an afternoon.

SECTION 09

Rules list & checklist screen

The statute is rendered inside the application as a table, not hidden inside the engine. An officer can read the rule, see what the system tested, and see the result — on one screen, in one row.

Reference table — the rules as published

Available at </rules> at all times, independent of any product. Searchable, filterable by rule chapter, and expandable to the full text of each sub-rule with its provisos.

RULE	SUBJECT	WHAT IT REQUIRES
6(1)	Declarations on every package	Manufacturer / packer / importer name and address, common or generic name, net quantity, month and year of manufacture or packing, retail sale price, dimensions where relevant.
6(2)	Consumer care details	Name, address, telephone number and e-mail, if available, of the person or office to contact for consumer complaints.
6(3)–(4)	Sticker restrictions	No sticker may alter a mandatory declaration; a revised lower MRP sticker is permitted but must not cover the original MRP.
7(1)–(3)	Principal display panel	Minimum numeral heights per Tables I and II; letters not below 1 mm, or 2 mm when blown, formed, moulded, embossed or perforated; width at least one third of height.
8(1)	Where declarations appear	All declarations on the principal display panel, with clear space around the quantity declaration — one numeral height above and below, twice that left and right.
9(1)–(4)	Manner of declaration	Legible and prominent; MRP and net-quantity numerals in a conspicuously contrasting colour; not readable only through liquid; in Hindi in Devanagari script or in English.
11	Quantity declaration	Net quantity excludes wrappers and packing materials; not to be qualified by “when packed” except as the Third Schedule allows.
12(2)–(6)	Manner of quantity	Correct unit class for the commodity; no exaggerated, misleading or inadequate expression such as minimum, average, about or approximately.
13	Units of measure	Gram below a kilogram, centimetre below a metre, millilitre below a litre, and the corresponding rules for area and volume.
18(2)	Sale above MRP	No person may sell a packaged commodity above the declared retail sale price.
24	Wholesale packages	Distinct declaration set for wholesale packages — identity, total number of retail packages, and net quantity.
31	Advertisements	Any advertisement quoting retail sale price must state net quantity, in the same font size as the price.
32	Penalty	₹4,000 for contravention of Rules 27 to 31; ₹2,000 for any other contravention not otherwise punishable.

SECTION 09 · CONTINUED

Live checklist — the same rules, applied to the open product

Rendered beside the extraction table on `/results/[id]`. Each row states what was found, what the rule demands, and the verdict. The officer column is editable, and any override requires a typed reason that is stored with the record. The example below is a 200 g biscuit pack with four failures and one item held for review.

RULE	CHECK	FOUND	REQUIRED	VERDICT
6(1)(a)	Manufacturer name & address	Present, 3 lines	Mandatory	PASS
6(1)(b)	Common / generic name	"Biscuits"	Mandatory	PASS
6(1)(c)	Net quantity	200 g	Standard unit	PASS
6(1)(d)	Month & year of packing	Not detected	Mandatory	FAIL
6(1)(e)	Retail sale price	MRP Rs 40 incl. of all taxes	Prescribed form	PASS
6(2)	Consumer care details	E-mail present, no telephone	Name, address, phone, e-mail	FAIL
7(2)	Numeral height	0.8 mm measured	1 mm min (up to 200 g)	FAIL
8(1)	Clear space around quantity	1.2× height at left	2× height left and right	FAIL
9(1)(b)	Contrast of MRP numerals	Ratio 3.1:1	Conspicuous contrast	REVIEW
12(6)	Misleading quantity wording	None found	No qualifiers	PASS
9(4)	Language of declarations	English	Hindi or English	PASS
6(1)(d)	Food exemption applies?	FSSAI category	Defers to FSS Rules	N/A

6

PASSED

4

FAILED

1

HELD FOR REVIEW

1

NOT APPLICABLE

₹8,000

INDICATIVE EXPOSURE

WHAT THE OFFICER SEES NEXT TO IT

Clicking any row opens the evidence crop for that check — the exact region of the label the verdict was drawn from, with the measured value overlaid. A failed font-height row shows the measured millimetre figure against the prescribed minimum, drawn on the numeral itself.

OVERRIDE DISCIPLINE

An officer can change any verdict, but the original machine finding is never erased. The row then shows both — what the engine concluded and what the officer decided — with the typed reason, the timestamp and the officer's identity attached.

The checklist is the report. What appears on this screen is exactly what is printed into the PDF. There is no second transcription step where a finding can drift from the evidence that produced it.

SECTION 10

Rule validation matrix

How each statutory requirement becomes a machine-testable predicate, and what the system does when it cannot decide. Automation level is stated honestly for every row — full, assisted, or flagged for human judgement.

CITATION	AUTOMATED CHECK	METHOD	AUTOMATION
6(1)(a)	Manufacturer / packer / importer present	Named-entity extraction over the address block; the qualifying words manufactured by, packed by and imported by are detected and attributed per the Explanations to Rule 6.	Full
6(1)(b)	Common or generic name	Classified against a commodity lexicon; brand name alone is rejected as insufficient.	Full
6(1)(c)	Net quantity declared	Regex plus unit parser over the panel; value and unit normalised and cross-checked against Rules 12 and 13.	Full
6(1)(d)	Month and year of packing	Date parser accepting words, numerals or both; category exemptions applied for food, cosmetics, bidi and LPG.	Full
6(1)(e)	Retail sale price	Pattern match on the prescribed wording; value parsed and rounding convention verified.	Full
6(2)	Consumer care details	Presence of contact name or office, address, telephone and e-mail; each component reported separately.	Full
6(3)	Sticker over MRP	Edge and texture discontinuity detection around the MRP region; officer confirms.	Assisted
7(2) T-I	Numeral height by weight / volume	Glyph height in mm from calibration, tested against the 1 / 2 / 4 mm bands, doubled for blown or embossed containers.	Assisted
7(2) T-II	Numeral height by length, area, number	Panel area estimated from calibration, then the 1 / 2 / 4 / 6 mm bands applied.	Assisted
7(3)	Letter height and width ratio	Minimum 1 mm letter height; width measured at no less than one third of height, excluding the exempted glyphs.	Assisted
8(1)	Panel placement and clear space	Panel classification plus geometric measurement of the printed-matter-free margin around the quantity declaration.	Assisted

SECTION 10 · CONTINUED

Rule validation matrix

CITATION	AUTOMATED CHECK	METHOD	AUTOMATION
9(1)(b)	Contrast of MRP and quantity	Local luminance contrast ratio between glyph pixels and background; ambiguous cases routed to review.	Assisted
9(2)	Not read through liquid	Transparency and refraction cues; flagged for the officer, never auto-failed.	Flagged
9(4)	Language	Script detection across the declaration block for Devanagari and Latin.	Full
11(2)–(3)	“When packed” qualifier	Phrase detection cross-referenced against the Third Schedule commodity list.	Full
12(2)	Correct unit class	Commodity state inferred, then mass, length, area, volume or number verified as the right basis.	Full
12(6)	Misleading quantity wording	Lexicon match for minimum, not less than, average, about, approximately and near-variants within the quantity region.	Full
13(2)	Unit sub-division	Gram below a kilogram, millilitre below a litre, centimetre below a metre.	Full
18(2)	Sale above MRP	Declared MRP compared against the shelf or listing price captured during inspection.	Assisted
31(2)	Advertisement font parity	For listing scans, net-quantity font size compared against retail-sale-price font size.	Assisted
–	Description spelling	Dictionary check with a domain lexicon; doubles as an OCR-quality signal.	Full

Assisted means assisted. Rows marked assisted or flagged produce a measurement and a provisional verdict, but the case does not close until an officer confirms it. The system is built to be defensible under challenge, which means never asserting more certainty than the image supports.

Confidence gating

Every check carries a confidence value derived from OCR token confidence, calibration quality and the margin by which a threshold was cleared. That value decides what happens to the row without any officer having to ask for it.

HIGH CONFIDENCE

The verdict is recorded as stated and shown in plain pass or fail styling. The officer can still override it, but nothing is demanded of them.

MEDIUM CONFIDENCE

The verdict is shown with a review flag and the row is highlighted. The case cannot be finalised until the officer has looked at that row explicitly.

LOW CONFIDENCE

No verdict is asserted. The row reads needs review, names what was missing — calibration, legibility, category — and the case joins the pending-review queue.

Why this matters more than accuracy. A compliance system that is right ninety-five per cent of the time and silent about which five per cent is unusable in enforcement. One that is right ninety per cent of the time and says clearly which ten per cent it is unsure about is entirely usable, because the uncertainty is routed to a person instead of buried in a verdict.

SECTION 11

End-to-end workflow

One inspection, start to finish. Steps 1 to 3 happen in the field; 4 to 6 happen in seconds on the server; 7 to 9 are the officer's.

1 Sign in and open an inspection

The officer authenticates, and the session carries their role and jurisdiction. A new inspection record is created with a reference number, timestamp and location. Offline, the record is queued locally and syncs when connectivity returns.

2 Choose camera or upload

From the side navigation the officer picks [Scan – Camera](#) to shoot the package on the spot, or [Scan – Upload](#) for images already taken. Barcode capture, where a barcode exists, links the scan to an existing product record.

3 Capture every panel and the scale reference

The principal display panel is mandatory; side and back panels are prompted for. A scale card or a typed package dimension is captured so millimetre measurement is possible. The client warns on blur, glare or a cropped panel before accepting the frame.

4 Automatic extraction

Images are conditioned, sent to the EasyOCR service, and the returned tokens are assembled into the six declaration heads. The results screen opens with both tables populated — raw OCR output on one side, structured declarations on the other.

5 Correct what the machine misread

Low-confidence cells are highlighted. The officer edits them in place; the spell checker flags suspect words in the description. Every correction is stored against the inspection, both as the authoritative value and as training data.

6 Rule evaluation

The rule engine runs the applicable pack against the corrected fields and the measured geometry. The checklist table fills with pass, fail, not-applicable or review for every rule, each carrying its citation and evidence crop.

7 Officer adjudication

The officer accepts the engine's findings or overrides any of them with a recorded reason, adds remarks, and attaches supporting photographs — JPEG or PNG, compressed to the 10–15 KB envelope on upload.

8 Generate and download the report

A compliance certificate or a violation report is produced as PDF, with DOCX and XLSX available alongside. The report is signed, carries a verification QR, and downloads directly from the results screen.

9 Commit, monitor, act

The inspection is written to MongoDB and appended to the product's compliance history. Dashboards update. Where a violation stands, the case moves into the enforcement track — notice, compounding, prosecution or closure — and is tracked to its outcome.

TYPICAL TIMING

Capture to populated checklist runs in roughly six to twelve seconds per package on a warm OCR service. The officer's review is the slow part, which is the correct place for the time to go.

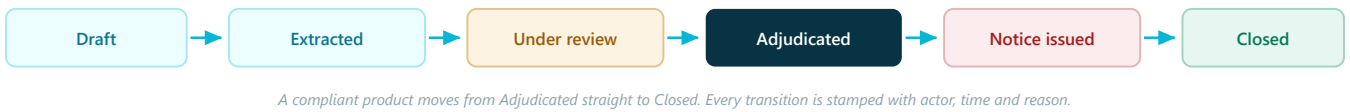
FAILURE BEHAVIOUR

If OCR times out, the inspection is preserved with its images and re-queued. If calibration is missing, the font checks report needs-review and the rest of the rules still evaluate. A partial result is never discarded.

SECTION 12

Roles, lifecycle & data model

Case lifecycle



Roles and permissions

ROLE	CAN DO	CANNOT DO
Field Inspector	Capture, upload, correct extraction, run checks, draft reports within their own jurisdiction.	Issue notices; edit rule packs; see other districts.
Senior Inspector	Everything above, plus adjudicate, override verdicts, issue and sign notices, reassign cases.	Edit rule packs; manage accounts.
Controller / Admin	Full jurisdictional oversight, user and role management, bulk exports, dashboard configuration.	Delete an adjudicated record — records are append-only.
Legal Officer	Author and version rule packs, review citations, approve amendments before they go live.	Adjudicate individual cases.
Auditor	Read-only access to every record, report and audit log, across jurisdictions.	Any write operation.
Manufacturer (optional portal)	Self-check own artwork before printing; view only their own submissions.	See enforcement data, other brands, or any inspection record.

MongoDB collections

COLLECTION	KEY FIELDS	COLLECTION	KEY FIELDS
products	gtin, brand, genericName, category, imageHash, declarations, complianceHistory[]	users	email, passwordHash, role, jurisdiction, mfaSecret, lastLogin
inspections	ref, officerId, geo, capturedAt, images[], ocrTokens[], extracted{}, results[], verdict, status	reports	inspectionId, format, fileId, hash, issuedBy, issuedAt, qrToken
rulepacks	version, effectiveFrom, rules[] {id, citation, applicability, predicate, severity, penalty}	auditlog	actor, action, target, before, after, ip, at — append-only, never updated

Append-only by design. An inspection that has been adjudicated is never mutated. Corrections create a new revision linked to the original, so the record of what was decided, when, and on what evidence stays intact — which is the difference between a database and an evidentiary record.

SECTION 13

Deployment & security

Technology stack

LAYER	CHOICE	NOTES
Front end	Next.js (App Router), React, TypeScript, Tailwind CSS	Server components for the dashboard and repository; client components for camera, upload and editable tables. Installable as a PWA for field use.
API	Node.js with Express, OpenAPI- documented	Multer for uploads with MIME sniffing; Zod for request validation; JWT sessions with refresh rotation.
Image processing	sharp, OpenCV bindings	De-skew, glare suppression, panel segmentation, and the JPEG/PNG compression that enforces the attachment size envelope.
OCR	EasyOCR in a Python FastAPI sidecar	English and Devanagari models; returns word text, bounding boxes and confidence. Scaled independently of the API, GPU-optional.
Rule engine	Node service over JSON rule packs	Pure functions, fully unit-tested, no network calls — the same inputs always produce the same verdict.
Spell check	nspell with Hunspell dictionaries	Indian-English base list extended with a curated commodity and brand lexicon.
Database	MongoDB with GridFS	Documents for inspections and products; GridFS for original captures; text indexes for search.
Queue	BullMQ on Redis	Bulk ZIP processing, listing crawls and report generation run asynchronously.
Reports	PDFKit / Puppeteer, docx, exceljs	PDF for the record, DOCX for editing, XLSX for analysis.
Deployment	Docker Compose for development; Kubernetes for production	Deployable to a state data centre or NIC cloud; CI/CD with automated rule-pack regression tests on every change.
Observability	Prometheus, Grafana, structured JSON logs	OCR latency, queue depth, per-rule failure rates and review-queue backlog as first-class metrics.

Security

IDENTITY

Password plus TOTP second factor for every officer account; short-lived JWTs with rotating refresh tokens; role and jurisdiction enforced server-side on every request, never in the browser.

DATA

TLS in transit, encryption at rest for the database and object store, SHA-256 hashes on every stored image, and an append-only audit log covering all reads of evidence as well as writes.

UPLOAD SAFETY

MIME sniffing rather than extension trust, size ceilings, image re-encoding to strip embedded payloads, and EXIF retained separately as metadata rather than passed through into the served file.

SECTION 13 · CONTINUED

Known limits & roadmap

Stated plainly, because a compliance tool that overstates its own certainty is worse than no tool at all. Everything below is a design position, not an outstanding defect.

Known limits

- **Font measurement depends on calibration.** Without a scale reference in frame, Rule 7 checks return needs-review. This is a deliberate refusal to guess, not a gap to be papered over.
- **OCR on curved, reflective or low-contrast packaging degrades.** The confidence score is surfaced rather than hidden, and poor captures are re-prompted at the camera rather than silently passed downstream.
- **Category exemptions are rule-driven and need legal upkeep.** Food and cosmetics defer to other statutes; those mappings are maintained by the Legal Officer role and versioned like any other rule.
- **The system reports; it does not adjudicate.** Penalty figures under Rule 32 are indicative exposure, and every enforcement action remains a human decision on the record.

Roadmap

NEXT

E-commerce listing crawler at scale, regional language OCR beyond Devanagari, and a manufacturer pre-print self-check portal.

THEN

On-device inference for offline extraction, barcode-linked national product master, and automated notice drafting from violation sets.

LATER

Anomaly detection across brands and districts to direct inspection effort where non-compliance is statistically concentrated.

Build sequence

PHASE	SCOPE	PROVES
1 — Skeleton	Next.js shell with side navigation, auth, MongoDB schemas, camera and upload pages.	An image can be captured, stored and retrieved against an inspection record.
2 — Reading	EasyOCR sidecar, image conditioning, both extraction tables with inline editing.	A label becomes structured, correctable data.
3 — Judging	Rule pack for Rules 6, 9, 12 and 13; checklist screen; spell check.	The text-based checks work end to end without calibration.
4 — Measuring	Scale calibration, Rule 7 font-height analysis, Rule 8 clear-space geometry.	The measurement-dependent rules, the hardest part.
5 — Reporting	PDF, DOCX and XLSX generation, attachment compression, evidence pack, QR verification.	A finding leaves the system as a usable document.
6 — Scaling	Dashboards, repository search, roles and audit, bulk and listing modes.	Many officers and many inspections, under oversight.

The shortest useful path. Phases 1 to 3 already produce a working compliance checker for the declarations that are simply present or absent — which is the majority of observed non-compliance. Phase 4 is where the system starts doing something an inspector cannot do quickly by hand.